

BLACKALL, AustraliaWMO: **953510**Lat: **24.4295S**Lon: **145.4306E**Elev: **925**StdP: **14.21**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.3	41.2	12.9	11.0	67.5	16.6	13.1	65.3	20.8	68.8	17.6	64.4	3.9	160	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.3	103.1	68.1	100.6	67.8	98.4	67.7	76.0	87.2	74.7	86.2	73.7	85.5	8.8	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.5	129.0	79.1	72.1	122.8	78.7	70.6	116.7	78.7	40.2	87.4	39.0	86.0	38.0	85.3	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.9	15.4	32.0	106.9	2.1	3.3	30.5	109.3	29.3	111.2	28.1	113.1	26.6	115.4		
(5)			29.0	77.3	2.4	1.7	27.2	78.5	25.8	79.5	24.5	80.5	22.7	81.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.0	84.6	82.7	80.4	73.6	65.3	59.8	58.9	61.8	69.4	76.1	80.8	83.6		
	DBStd	10.79	4.83	4.67	4.69	4.94	5.52	5.50	5.49	5.97	6.33	6.08	5.30	5.32		
	HDD50	4	0	0	0	0	0	1	2	1	0	0	0	0		
	HDD65	611	0	0	0	4	62	173	202	137	28	5	0	0		
	CDD50	8408	1074	915	942	709	475	293	277	365	581	810	924	1043		
	CDD65	3540	609	495	477	263	72	16	12	36	158	350	474	578		
	CDH74	43357	8032	6167	5573	2675	687	137	137	520	2013	4155	5798	7463		
	CDH80	22395	4607	3360	2822	1034	123	7	10	111	814	2077	3165	4265		
(14) Wind	WSAvg	7.5	8.1	7.9	7.6	7.0	6.7	6.6	6.7	7.2	7.6	8.4	8.5	8.3		
(15) Precipitation	PrecAvg	20.2	3.5	2.8	2.2	1.4	1.4	0.7	0.8	0.7	0.6	1.1	2.0	2.7		
	PrecMax	31.0	15.6	9.1	14.5	11.8	9.4	3.9	5.0	4.7	2.5	4.8	8.0	9.7		
	PrecMin	10.2	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1		
	PrecStd	6.3	3.3	2.2	2.8	2.0	1.7	0.9	1.2	0.9	0.8	0.9	1.9	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.6	105.4	101.5	94.1	87.3	80.9	81.4	87.6	96.4	100.6	103.6	105.7		
		MCWB	68.6	68.8	68.6	68.5	63.6	61.3	61.3	59.8	62.3	64.3	66.8	67.3		
	2%	DB	102.3	101.0	97.8	91.0	83.2	77.9	77.7	83.0	91.4	96.6	100.2	102.4		
		MCWB	69.2	68.6	68.7	65.8	62.2	60.2	58.7	59.0	61.1	63.5	65.9	67.7		
	5%	DB	99.6	97.5	94.8	88.4	80.7	75.3	75.0	79.6	88.1	93.7	97.2	99.4		
		MCWB	69.4	69.3	68.4	64.4	60.9	59.0	57.3	57.5	60.4	63.4	65.9	67.1		
	10%	DB	96.8	94.5	92.1	85.9	78.1	72.3	72.2	76.4	84.9	90.5	94.1	96.4		
		MCWB	69.3	69.4	68.0	63.6	59.7	57.2	55.9	59.7	62.4	65.1	67.3	67.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.7	77.3	76.2	72.1	69.3	64.8	66.3	64.9	69.2	70.7	73.7	76.5		
		MCDB	90.0	88.6	84.1	83.8	76.6	75.5	74.1	78.7	80.6	86.3	84.2	88.3		
	2%	WB	75.9	75.8	73.8	70.3	66.0	62.7	63.2	62.0	66.7	68.8	71.4	74.5		
		MCDB	87.8	86.5	83.6	83.1	76.6	71.9	71.3	76.4	78.2	84.0	83.7	87.8		
	5%	WB	74.6	74.7	72.6	68.7	64.0	61.2	60.6	59.9	64.4	67.4	70.1	73.1		
		MCDB	86.5	85.5	84.1	81.6	76.0	70.6	70.1	74.6	78.3	83.2	84.3	85.9		
	10%	WB	73.6	73.6	71.4	67.0	62.3	59.5	58.2	58.0	62.4	66.1	68.9	71.9		
		MCDB	85.8	85.4	84.9	79.7	74.2	69.0	69.1	72.7	79.5	81.6	84.4	85.2		
(35) Mean Daily Temperature Range	MDBR		21.3	21.5	22.0	24.0	24.7	24.1	25.9	28.3	28.1	26.1	23.5	21.9		
	5% DB	MCDBR	25.3	25.7	25.0	25.9	26.3	25.8	28.5	30.8	31.4	29.1	27.4	26.2		
		MCWBR	8.0	7.5	8.0	10.0	11.4	12.0	14.0	14.3	12.9	11.0	8.7	8.7		
	5% WB	MCDBR	19.2	19.0	18.6	20.9	19.7	19.7	21.4	25.4	23.4	23.1	20.7	19.9		
		MCWBR	6.8	6.8	6.6	8.5	9.4	9.6	11.3	12.4	10.4	9.3	7.4	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.373	0.362	0.342	0.327	0.309	0.297	0.290	0.306	0.344	0.361	0.377	0.379		
	taud		2.505	2.541	2.581	2.575	2.570	2.604	2.602	2.529	2.432	2.417	2.420	2.454		
	Ebn at Noon		308	307	304	295	288	285	292	298	298	303	305	306		
	Edh at Noon		36	35	31	29	27	25	25	30	36	38	39	38		
(44) All-Sky Solar Radiation	RadAvg		2311	2209	1987	1710	1398	1213	1338	1658	1967	2259	2335	2358		
	RadStd		152	153	131	118	68	78	89	84	138	159	179	171		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page